FELIX

Second Engineering Systems Report

Integrated v3: progress against the September 22-24 baseline, the September 29 strategic-intelligence update, and the adaptive-execution design decision

PRIMARY FINDING

Felix has a functioning governance harness, a provenance-aware capability supply chain, an advisory router, and usage telemetry. It still has not joined them into an independently operated task-to-delivered-outcome control loop. The September 29 external findings strengthen the case for an independent qualification and outcome-evidence layer. Adaptive execution should extend native agent delegation across authorized compute pools rather than create a second orchestrator. Merge is not the endpoint.

REPORT BASIS	VALUE
Frozen code window	c6989dc5168f → 5772afac9357 | September 24-26, 2026 | endpoint September 26, 14:36 PDT
Operational telemetry	September 24, 00:40:21 - September 26, 00:00:00 PDT (exclusive end)
Strategic update	External intelligence note issued September 29, 2026; no new repo/telemetry claims implied
Report issued	September 29, 2026
Model-routing update	Claude Sonnet 5.5 / Opus 5.5 external benchmark review added September 30, 2026; routing guidance is provisional and not a shipped Felix capability.

Internal, evidence-qualified assessment. Engineering metrics remain bound to the frozen September 26 endpoint. September 29 additions are strategic/design updates unless explicitly identified as observed evidence.

Report map

- 1. Executive synthesis and decision
- 2. Measurement boundaries and comparison discipline
- 3. Execution scale, human leverage, and accepted output
- 4. Verification economics and the CI constraint
- 5. Trust Plane: operational and unfinished controls
- 6. Capability Plane: discovery, telemetry, adaptive execution and provider federation
- 7. Landing authority and the missing independent actor
- 8. Product thesis, external strategic implications and evidence packet
- 9. Revised minimum viable closed loop
- 10. Sequenced roadmap, validation and pilot criteria
- 11. Founder attention and the control interface
- 12. Competitive boundary and enterprise system-of-record implications
- 13. Commercial validation experiments and decision gates
- 14. Risks, uncertainties and next measurement
- Appendix A. Metrics and baseline comparison
- Appendix B. Evidence and source register
- Appendix C. Reusable report skeleton and self-generation trigger
- Appendix D. Adaptive execution design annex

READ THIS FIRST

The September 27 v2 report remains the engineering baseline. This integrated edition does not silently update its code or telemetry window. It adds the September 29 strategic research and the adaptive-execution design decisions discussed after v2, while preserving the original comparison discipline.

1. Executive synthesis and decision

The frozen engineering window shows sustained large-scale multi-agent execution, continued verification investment, and meaningful improvements in checker qualification, fail-closed parsing, and GitHub gate reuse. The integration frontier remains: qualified evidence, candidate landing, controlled worker dispatch, and delivered-state assurance must become one operational loop rather than independent subsystems.

Finding	What the evidence establishes
Accepted output	32 first-parent commits, 69 files, +6,833/-722 lines. Test additions were 3,564 against 1,070 engine lib/CLI additions: a test-heavy hardening interval, not an equivalent feature sprint.
Operating scale	25,988 distinct model responses and 27,386 tool calls; 5.673B raw token traffic, 96.78% cache read. Subagents carried 58.82% of traffic.
Human interface	216 parent sessions and 498 human text turns; 80.56% of sessions had at most one such turn. This omits UI approvals and other human actions.
Trust gap	In the bounded native audit, 46 of 88 recognized edit-ending stops had stale or no recognized verification. Most nominal passes had obscured command-status semantics.
Landing bottleneck	34 PRs created, 33 merged within the code window, but 19 still open at freeze, 12 drafts. The landed merge command is not the planned independent landing actor.
Strategic signal	The September 29 note reports legal uncertainty around autonomous-agent actions and an incumbent governed-agent launch inside a system of record. This supports the problem framing, not a claim of market size or willingness to pay.
New capability decision	Provider federation is infrastructure. Adaptive execution is the feature: extend native delegation across authorized models, subscriptions and API pools while keeping Trust Plane acceptance independent of model choice.

DECISION FOR THE NEXT CYCLE

Run two concurrent engineering tracks: (A) harden independent acceptance and landing authority; (B) demonstrate a bounded Capability-to-Trust vertical slice that includes a narrow adaptive-execution path. In parallel, run explicit commercial/evidence experiments. Do not wait to perfect the Trust Plane, do not rebuild commodity runtime/identity infrastructure, and do not grant unrestricted autonomy merely to accelerate the demonstration.

2. Measurement boundaries and comparison discipline

The accepted-code comparison remains the frozen Git range c6989dc5168f5113745591fde1115b7270dfe933 → 5772afac9357831f61a3bb60542106abf7523a4f. Its 32 first-parent commits are landed results. Working-tree experiments, unmerged pull requests, retrospective telemetry, and all design discussion after the endpoint are not credited as accepted output.

The active telemetry ends at midnight PDT on September 26, while Git history freezes at 14:36 PDT that afternoon. Those are deliberately different windows. The prior report used the preceding 48-hour canonical code window; its operating counters had some near-window sampling. Comparison is descriptive, not a synchronized intervention study.

The September 29 material is an analytical overlay. External findings are source-qualified research; adaptive execution/provider federation is a proposed Capability Plane direction. Neither is represented as shipped at the frozen SHA.

Method register: Git first-parent history and diffstat; response-ID-deduplicated Claude Code JSONL; parent-session text-turn analysis; native felix audit on filtered parent transcripts; GitHub Actions attempt/job records; frozen source inspection; September 27 v2 report; September 29 strategic-intelligence note; this session's architecture decisions.

- Raw token traffic includes cache reads, cache creation and output. It is not fresh inference tokens, API spend, or a productivity measure.
- Parent-session audit cannot observe subagent edit/verification events. Its check roster and write recognition are bounded.
- GitHub Actions job elapsed minutes are not billed runner minutes; skipped jobs can have zero or invalid elapsed timestamps.
- Merged-PR latency excludes PRs still open at freeze and is therefore right-censored.
- Implementation claims come from code paths at the frozen SHA. Historical roadmap, source comments, and this report's proposed architecture are not automatically current state.
- External-news claims are carried from the attached September 29 research note and retain its FACT/HYPOTHESIS/UNKNOWN distinctions.

3. Execution scale, human leverage, and accepted output

3.1 Model and tool execution

Metric	New window	Previous report
Unique responses	25,988 (7,152 parent; 18,836 subagent)	30,694
Raw token traffic	5.673B (5.490B cache read)	6.975B
Cache-read share	96.78%	97.32%
Subagent traffic	3.337B; 58.82%	3.728B; 53.45%
Tool calls	27,386; Bash 21,623 (79.0%)	32,187; Bash 26,295 (81.7%)
SendMessage events	553	Not comparable / not stated

These quantities confirm a distributed execution topology with substantial context reuse. They do not establish proportional improvement in delivered software or autonomous decision quality. The new window contains fewer accepted commits and fewer raw tokens than the prior burst; the mix of work changed materially, making simple output-per-token interpretations unsafe.

3.2 Supervision concentration

The bounded scan found 216 active parent sessions with 498 human text turns. Forty sessions had zero, and 134 had exactly one; 174/216 (80.56%) had at most one, and 186/216 (86.11%) had at most three. One session contained 57 turns, so the mean conceals a substantial supervisory tail.

3.3 Accepted output and verification coupling

The canonical range contains 32 first-parent commits, 69 changed files, 6,833 additions and 722 deletions (net +6,111). The inspected category allocation found 3,564 test additions, 1,070 engine library/CLI additions, 1,763 documentation additions and 288 CI additions. Test additions exceeded engine library/CLI additions by roughly 3.33:1.

The prior interval had 50 commits, 24,072 additions and 6,929 deletions, with 6,060 test and 5,995 engine additions. This period reads primarily as qualification, fail-closed behavior, security and CI-economics hardening rather than another broad architectural expansion.

4. Verification economics and the CI constraint

4.1 Native audit: the key negative signal

The exact active-window audit covered 216 sessions and 88 recognized stop-after-edit events. It classified 42 nominal passes, 29 stale, 17 never checked and zero explicit failed endings. Thus 46/88 (52.3%) had no fresh recognized verification. Forty-one of the 42 nominal passes had command/status-obscuring shapes, so "passed" means the audit recognized a nominal success, not that the underlying verification status was independently established.

Felix emitted 371 Stop holds across 45 sessions: 64 after counted edits, 54 after excluded scratch/config edits, and 253 after no new edit. The earlier report counted 363 holds and a 35.7% stale/never edit-stop share under a different scan. Do not interpret the share change as degradation or improvement without a stable audit instrument.

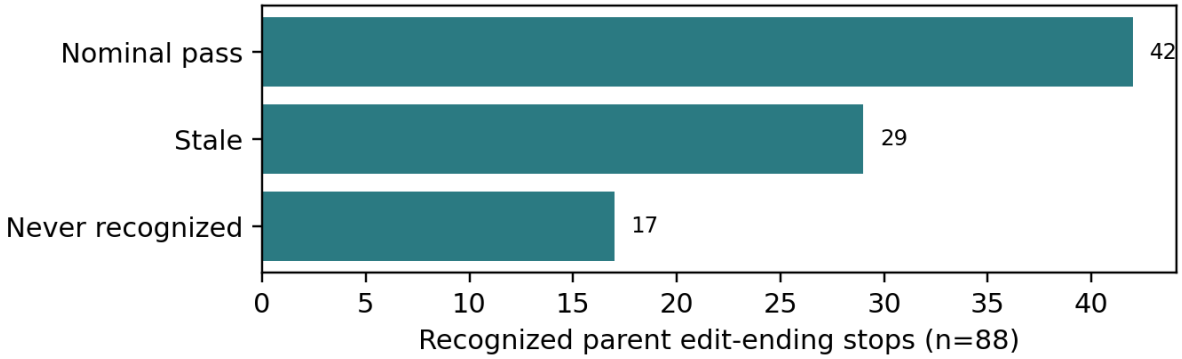


Figure 1. Native parent audit: 46 of 88 recognized edit-ending stops lacked fresh recognized verification. Of 42 nominal passes, 41 carried command/status ambiguity.

INTERPRETATION

The dominant risk is evidence freshness and truthfulness, not an agent explicitly ending immediately after a visible red check. A worker can report completion when an old check no longer describes the candidate or a shell command hides the real status. This directly requires authoritative, revision-bound evidence.

4.2 Remote verification capacity

For September 24-25, 113 gate workflow run IDs yielded 117 attempts, including reruns. Across 174 job records, positive elapsed time summed to 606.27 job-minutes (not billed minutes). Eighty successful full-gate jobs consumed 516.18 minutes; 19 failed jobs consumed 86.68; one cancelled job consumed 1.38; and 17 successful reuse jobs consumed 2.02 minutes.

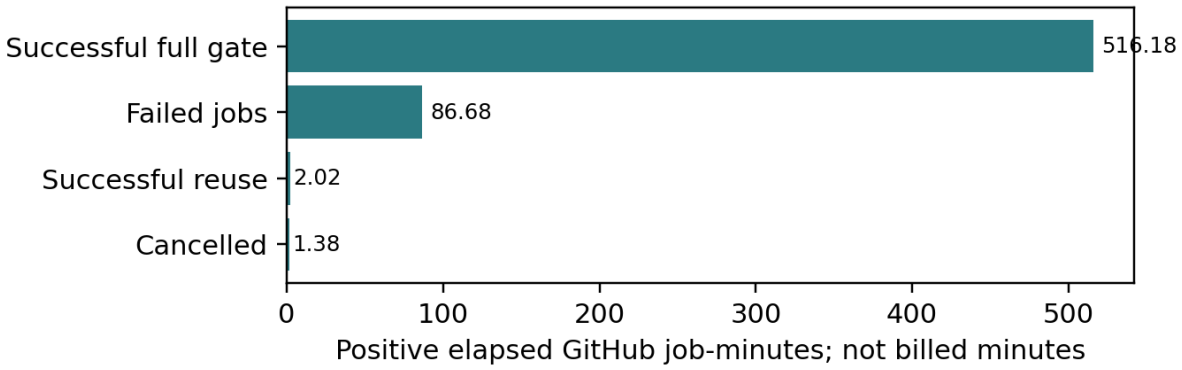


Figure 2. GitHub Actions verification work in the active CI window; durations are summed positive job elapsed time, not billed runner minutes.

4.3 Capacity needs to be scheduled

Earlier operation exhausted roughly 2,000 included GitHub Actions minutes in about four days. Once canonical landing requires remote evidence, verification capacity can halt landing while workers continue producing candidates.

- Tier checks: affected local tests first; reserve full remote CI for landing-frontier candidates.
- Invalidate only evidence affected by changed inputs; reuse exact qualified trees where authority and environment permit.
- Cancel superseded workflows; prefer queue-aware admission of candidates to unbounded green-test attempts.
- Display verification budget, evidence freshness, queue age and reason for hold in the control interface.

5. Trust Plane: operational and unfinished controls

Mechanism	Observed status at frozen SHA
Stop hold / receipts	Daily use; enforcement exists, but parent audit shows many stale or unrecognized check patterns.
Gate qualification	If qualification.tsv exists, gate runs felix qualify --gate and nonzero fails the stage. Missing table is announced and skipped.
Six-control qualification	Declared controls are exercised; shortfall in the declared six is reported, not universally refused.
Obligations	Declared active blocking/UNKNOWN obligations fail gate. Authorized exceptions are distinguished from discharged obligations.
Capability needed by obligation	Checker IDs resolve against qualification.tsv but are report-only and do not change discharge verdicts.
Commissioning	Empty/malformed table fails gate; stale/unknown receipt is announced there and is stricter at merge.
Judge identity	Self-governed felix merge checks an installed byte-identical main-accepted judge; gate not fully pinned.
Unattended repair / exception	Built or represented, but no validated real unattended repair and no granted exception at snapshot.

The key missing join is not a generic absence of tests. A required obligation must remain UNKNOWN until a checker that satisfied the complete required control set proves discharge for the precise candidate and applicable environment. The roadmap raises qualification to seven controls and makes independent checker evidence a prerequisite. That is planned, not armed at the frozen endpoint.

There is a temporal authority problem as well: the worker currently runs checks inside its own session, while merge has a stronger accepted-judge path. A complete Trust Plane must decide what the worker may write, which checker may decide, which revision was checked, and which external facts invalidate the result.

5.1 September 29 extension: outcome evidence

The external research sharpens the Trust Plane objective from "tests passed" to a durable acceptance record: what was authorized, what requirements applied, what actually happened, what independently established acceptance, and which later changes can invalidate that acceptance.

AGENT OUTCOME EVIDENCE PACKET

Proposed packet fields: (1) task authority; (2) execution boundary; (3) execution provenance; (4) qualification evidence, omissions and uncertainty; (5) acceptance decision and exceptions; (6) delivered-state evidence, including integration, deployment, rollback or monitoring state that affects continuing acceptance.

Integrity of an audit record is not proof that task requirements were complete or behavior correct. Every packet must state evidence coverage and UNKNOWNs.

6. Capability Plane: discovery, telemetry and adaptive execution

6.1 What is already substantial

The capability supply chain is more developed than a superficial "dispatch not built" description suggests. Felix discovers project capabilities, records adoption, generates setup obligations, accepts provider nominations as INFERRED, fetches pinned source, inspects shipped surfaces, derives OBSERVED provenance and risk, and applies a centralized admission policy before declaration or installation.

Risk	Person-declared	Engine-observed	Model-inferred
Low	Install	Install	Hold
Medium	Install	Hold	Hold
High	Refuse	Refuse	Refuse
Unknown tier/source	Refuse	Refuse	Refuse

A declaration is not installation. felix stack plan uses felix_admit and stack apply --yes installs only plan actions marked install. The commissioning run checks environmental state and provenance, re-reads after manifest changes, and writes receipts against post-write state. Legacy stack rows without a grounding column default to DECLARED, which needs explicit migration treatment in a stronger contract.

6.2 Routing and observable use

Routing scores typed intent and repository context, renders a play into agent context, records misses and declines, and recognizes machine-generated messages to avoid treating them as human tasks. It is intentionally advisory; a fallible keyword match is not an execution grant. The ledger records tool signals after tool use, folds at SessionEnd and can catch up stranded session records.

Retirement currently starts from capabilities named by a route and never observed reached across a threshold of sessions. Hook-only plugins are explicitly unobservable, mixed hook/tool plugins are classified separately, and unrouted entries are not automatically treated as unnecessary. This is usage telemetry rather than measured task success or capability qualification.

6.3 Where the plane stops

No shipped operational felix dispatch entry point or landing actor was found in the frozen file/entry-point inventory. The intended dry run only prints an execution lease; it launches nothing. Sandboxing, temporary tool grants and revocation beyond session termination are absent. The current host is Claude Code only. These gaps make the current plane an admitted/routed capability system, not an independent engineering execution authority.

FOUNDATIONAL REUSE

Do not replace the existing catalogue, commissioning, pinned-source inspection, routing and ledger with a second registry. Promote their outputs into a task-bound capability contract: need, provider, qualification evidence, lease, measured outcome, freshness, resource pool and retirement rationale.

6.4 New design decision: native delegation first

Claude Code already performs part of the desired orchestration: the main agent can decompose work, spawn subagents and assign different model classes to different workers. Felix should not build a redundant planner that second-guesses every delegation. The main agent remains the primary task decomposer and first-pass difficulty estimator; Felix governs eligibility, authorization, resource pools, evidence, budget and acceptance.

Responsibility	Primary owner
Interpret objective and decompose work	Main agent / native agent runtime
First-pass task difficulty estimate	Main agent, informed by Felix evidence
Choose among eligible worker classes	Native runtime within Felix policy
Cross-provider execution and authorized resource pools	Felix provider adapters
Permissions, budgets, context handoff and attribution	Felix Capability Plane
Qualification and completion authority	Felix Trust Plane
Escalation after contradictory or failed evidence	Main agent + Felix runtime evidence

6.5 Adaptive execution: the actual feature

Routing should be treated as two coupled decisions, not two separate products. (1) Within the currently selected host/worker, Felix should eventually help govern model class and reasoning effort (for example, Claude Sonnet versus Opus, including main-agent and subagent choices where the host permits it). (2) Once multiple authorized compute pools exist, Felix should also choose among workers/providers and then the model/effort configuration available inside that pool. The long-run routing object is therefore the whole execution configuration - worker/provider, model, effort, context and parallelism - while Trust requirements remain invariant.

This report does not settle the mechanism for making that joint allocation decision. Whether it should be rule-based, model-judged, learned from execution history, or hybrid remains an architecture question for a dedicated Claude/parallel-agent brainstorm and empirical prototype.

Cross-provider routing is infrastructure. The difficult feature is recognizing the minimum sufficient intelligence for the current work unit, then adapting when execution evidence changes the estimate. The system should support both escalation and de-escalation rather than allowing one difficult subproblem to pin an entire session to the most expensive configuration.

Signal	Routing implication
Established, repeatable operation	Prefer lower-cost eligible worker.
Unfamiliar subsystem or high ambiguity	Increase reasoning allocation or use a stronger worker.
Repeated repair attempts / contradictory evidence	Escalate, revise the plan or request independent diagnosis.
Hard problem resolved; remaining work routine	De-escalate back to standard/economy execution.
High consequence / low reversibility	Keep execution and acceptance distinct; strengthen independent verification even if implementation is simple.
Measured success on comparable tasks	Allow empirical policy to lower expected-cost allocation over time.

Difficulty, consequence and reversibility must remain separate variables. A simple database edit can be operationally consequential; a difficult algorithm in an isolated prototype can be reasoning-heavy but low-authority. Model expense is never a substitute for qualification.

Expected-cost objective

> **ROUTING OBJECTIVE**
>
> Choose the minimum sufficient eligible configuration by expected total cost, not per-token price: E[C] = C_execution + P(failure) x C_recovery + C_verification + C_switching. Include prompt/cache reconstruction, retries, handoffs and quota/CI pressure. A cheap worker that repeatedly fails can cost more than a frontier worker used once.

6.6 Main-agent adaptation

Subagent routing alone does not solve the case where the expensive main session remains active after the hard work is finished. Where the host supports model/effort changes between turns, Felix can govern those transitions at explicit checkpoints. Where it does not, Felix should continue through a new worker using durable state. Do not assume in-flight hot-swapping of a model during one generation or transfer of hidden reasoning across model families.

6.7 Provider federation

Felix should expose every user-authorized compute pool behind one dispatch contract: subscription-backed Claude Code, subscription-backed Codex/OpenAI execution where officially supported, API-credit pools, and later providers. Specific model catalog names should be discovered at runtime rather than hard-coded into policy.

Resource type	How Felix should treat it
Native host subscription	Primary local execution pool; quota/availability tracked where observable.
Second provider subscription	Optional execution pool enabled by explicit user toggle and supported provider authentication.
API credits / pay-as-you-go	Separate pool with explicit spend policy and cap; never silently substituted for subscription usage.
Additional providers	Adapter interface only after capability/authentication/permission semantics are explicit.

Credentials must stay outside agent-controlled worktrees and logs. Use provider-supported authentication and user-authorized keys only; no credential harvesting, scraped shared keys, or opaque third-party key pools.

6.8 Context continuity across providers

The handoff should not dump an entire Claude transcript into Codex or another provider. Felix should make engineering state durable outside hidden model context: task objective, accepted decisions, relevant revision, files, completed work, unresolved

questions, evidence, permissions and next action. Workers inspect the repository directly within an isolated worktree or equivalent bounded environment.

TASK-SCOPED CHECKPOINT

Minimum checkpoint: task ID, parent objective, exact revision, current stage, requirements/obligations, completed steps, unresolved items, evidence references, next action, allowed tools/paths, selected resource pool and compute tier. The artifact - not hidden reasoning - becomes transferable institutional memory.

6.9 User control and resource policy

Control	Example policy
Routing mode	Adaptive | preferred provider with fallback | manual only
Connected pools	Claude subscription | Codex/OpenAI subscription | provider API credits | other adapters
Paid fallback	Never | require approval | allow within explicit cap
Exhaustion behavior	Fallback to another eligible pool, reduce concurrency, or hold for user authority
Evidence policy	Never lower qualification because a cheaper or subscription-backed model was used

7. Landing authority and the missing independent actor

The landed felix merge command is a substantive attended boundary. Its blocker function checks exact-tree receipt, evidence gaps, probes, release obligations and commissioning conditions, with missing/empty declaration tables refused at merge. Self-governed merges refuse an environment-selected judge and compare the installed judge to the accepted main engine. The command verifies branch/commit/local tree again before invoking a GitHub squash merge.

However, it uses the branch as the final GitHub selector, and the post-merge ledger write is best effort. The frozen implementation inventory found no land.tsv, cmd_inventory, cmd_land or dedicated landing actor. Open PRs #350, #358 and #383 were not counted as landed. These remain distinct from the existing merge command.

Attended merge today	Independent landing actor requires
One checked-out candidate	Complete inventory: local/unpushed/orphan work plus PRs and blocked sessions
Current local evidence	Remote revision identity and exact evidence binding
Operator invokes felix merge	Eligibility queue, scheduling, explicit grant and order
Best-effort postmerge record	Durable result/reconciliation for the actual landed revision
Merge acknowledgment	Post-deploy evidence, rollback/migration obligations and production health

At the code freeze, 34 PRs were created and 33 merged within the Git window; median latency among merged PRs was 4.51 hours. Nineteen PRs were open, twelve drafts. Candidate generation is not equivalent to reconciled delivery.

8. Product thesis: plain language, technical formulation and strategic implication

8.1 Plain-language account

WHAT FELIX IS SUPPOSED TO DO

You describe the outcome. Felix works out what the project needs, finds and assigns suitable workers, checks their work using rules the workers cannot quietly rewrite, remembers what remains unresolved, and asks you only for decisions that actually require your authority. Work is not complete merely because code was merged; Felix must establish that the delivered result behaves as required.

8.2 Two interdependent planes

The Capability Plane answers: what must be done, what tools and workers can do it, who should be assigned, when they should run, how much compute they need, how well they performed and when a capability should retire. The Trust Plane answers: what is known versus UNKNOWN, which obligations bind, who proves which claim, against what revision and external facts, and who may authorize each state transition.

Capability without independent acceptance can generate large volumes of untrusted work. Trust without execution selection and feedback becomes a strong harness around somebody else's orchestration. Felix's claimed product is the feedback loop between them, not either plane in isolation.

8.3 September 29 strategic implication: authorization is not acceptance

The attached strategic note identifies an opening between authorization to act and acceptance of an outcome. Its Reuters-based finding is that autonomous-agent actions can create uncertain legal/economic exposure even when agents have deep customer-system access; its Nasdaq finding is that an incumbent is placing governed agents inside a system of record. These are problem and architecture signals, not proof of a new market or Felix demand.

The resulting product thesis is narrower than a generic agent-control plane: Felix should become an independent engineering-qualification and outcome-evidence layer - binding task-specific authorization, versioned obligations, independent evidence, acceptance and later invalidation - while integrating established runtime, identity, sandboxing and telemetry systems rather than reproducing them.

9. Revised minimum viable closed loop

9.1 First: restricted provider-answer lease

Retain the September 19 v0 boundary: one human-requested tick; a bare restricted session; strict MCP configuration; no plugin, Bash or settings source; explicit budget; and exactly one allowed answer-file edit. It emits a lease, a result and a human-readable hold, then enters Trust through the provider protocol. A person still commits admitted material.

The earlier launcher comparison (default: 27 plugins, 12 MCP servers, 434 skills; reduced launch: one plugin, zero MCP, 17 skills) is not validation of the final zero-plugin specification. Read permission under --restricted with dontAsk is explicitly unestablished. Revocation means ending the session, not dynamic revocation within it.

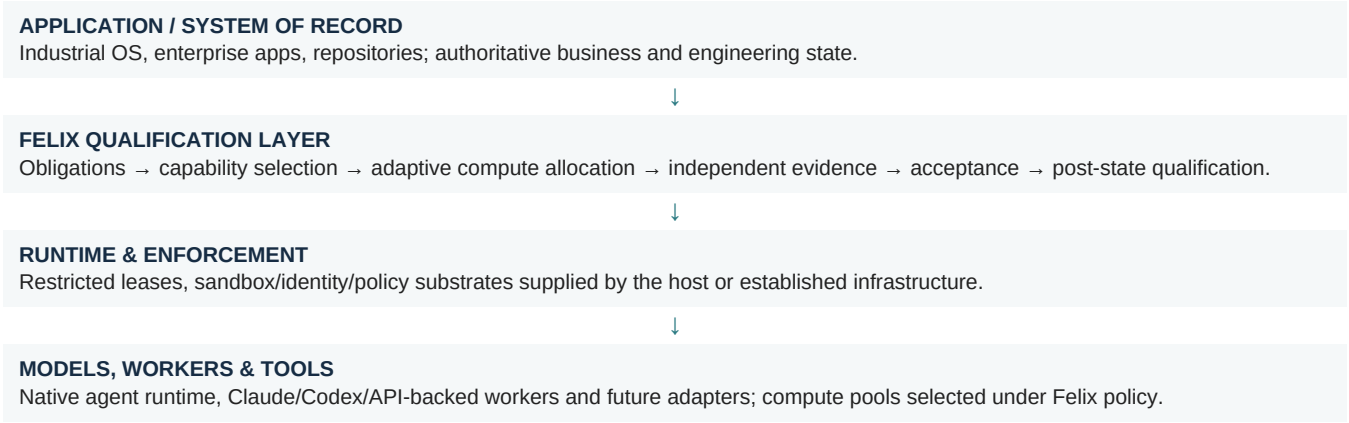
9.2 Then: engineering vertical slice

1. Capture one real objective and its requirements from user intent, repo state, domain rules and independent omission challenges. Keep unresolved obligations UNKNOWN.
2. Choose one eligible provider/worker by a task-bound qualification contract. Native delegation may select among eligible workers; Felix supplies resource-pool, permission and evidence constraints.
3. Issue a bounded engineering lease, preferably attended initially, with explicit paths, tools, compute/spend budget and exit conditions.
4. Collect attributable outputs and evidence; independently verify every blocking obligation against the exact candidate and necessary external facts.
5. Authorize one landing through the stronger accepted-judge and remote-revision boundary, then reconcile the actual landed state.
6. Observe a real delivered-state property, define a rollback/escalation path and preserve evidence that can go stale after merge.
7. Run one controlled adaptive-execution transition: either escalate after demonstrated difficulty or de-escalate after the hard portion is resolved, without changing the Trust acceptance bar.

9.3 Completion criteria

Dimension	Minimum evidence for the demonstration
Correctness	Seeded known-pass and known-fail cases; independent verdicts; no false green on unqualified/unknown claim.
Authority	Worker cannot alter obligations, judge selection or its own grant; admissions and landing remain authorized.
Reproducibility	Candidate and environment referenced precisely; evidence can be replayed or visibly marked not reproducible.
Utility	Time to usable result, founder interventions, verification cost and escaped defects compared against plain Claude Code.
Portability	Demonstrate on Felix and a materially different ecosystem, including Elenchus where feasible.
Adaptive compute	At least one measured routing decision has a documented baseline, cost, repair count and final accepted quality.
After merge	At least one post-merge/deploy check and a defined failure response are actually exercised.

9.4 Reference architecture



The Capability Plane allocates models, tools and compute against task uncertainty and requirements. The Trust Plane establishes whether work and delivered state meet versioned obligations. Merge is not the endpoint: deployment changes, migrations, dependencies and rollback evidence may reopen acceptance.

10. Sequenced roadmap, validation and pilot criteria

10.1 Connected tracks

Track	Near-term sequence	Do not claim until
A. Trust / landing	Inventory/reporting → shadow actor → attended landing → judge pin on gate/checkers → seven controls + obligation join → evidence staleness/postdeploy.	Shadow agreements, attended landings and authorization evidence pass.
B. Capability / adaptive execution	Seeded-omission test → install-keyed retirement → dry-run lease → real provider-answer tick → provider adapter/federation → qualified worker experiment → measured adaptive escalation/de-escalation.	Real restricted launch, correct records, independent admission and at least one measured routing transition are observed.
C. Validation / market evidence	Outcome-evidence review test → authorized cross-boundary handoff → post-merge invalidation test → compute-routing economics test.	Independent evidence materially improves a concrete workflow after overhead; buyer problem and repeatability are demonstrated.

The committed roadmap places the qualified judge in November, whole-life evidence in December/January, public replay in January/February and operational dispatch no earlier than March. Preserve safety dependencies, but evaluate whether a limited attended vertical slice can establish product value earlier. Do not make the Capability lane merely a spare quarter of sessions while the system claims to be dual-plane.

10.2 Adaptive execution sequencing proposal

The sequence in this report is a prioritization proposal, not a settled implementation architecture. It identifies what should be learned or made available early because of product closure and recursive-development ROI. Credential isolation, context transfer, subscription-backed execution, provider failure semantics, cache preservation, dispatch internals and the final router mechanism should be resolved later through dedicated high-capability model brainstorming, adversarial review and measured prototypes.

Stage	Scope	Exit criterion
1. Dispatch contract	Bounded worker receives revision, task, tools, paths, budget and exit conditions.	One real restricted launch and attributable result.
2. Provider adapters	Expose authorized Claude/Codex/API-backed compute behind one contract.	Same bounded task launches correctly through at least two eligible pools.
3. Passive routing	Record proposed model/pool decisions without changing execution.	Calibration dataset exists with actual outcomes and costs.
4. Adaptive delegation	Allow native main agent + Felix policy to choose among eligible workers.	Observed quality does not degrade versus fixed-policy baseline.
5. Escalation / de-escalation	Change worker/compute tier at checkpoint based on evidence.	At least one successful transition with preserved state and independent acceptance.
6. Empirical optimization	Use historical outcomes to minimize expected total cost under quality/authority constraints.	Policy beats frontier-only and naive cheap-first baselines on predeclared metrics.

10.3 Prerequisites and stopping rules

- Fix decision-epoch portability across PATH environments before scheduling discovery; the README still documents this staleness risk.
- Filter newest-workflow logic by gate workflow before another workflow can distort auto-merge streak or repair budget.
- Demonstrate provider requests end to end on real projects; proposal counts without admissions do not establish model-derived completeness.
- Never load untrusted repository settings into a credential-bearing headless runner; validate actual restricted Read/Write behavior.
- If seeded-omission recall does not beat plain Claude Code, narrow the claim to human-declared obligations and disclose the result.
- If adaptive routing saves nominal model cost but increases repair, context reconstruction, verification or escaped defects, it has failed.
- If a checker cannot fail on the Elenchus known-bad revision and pass on its corrected head, revise the control set before arming it.
- If clean-machine replay disagrees, do not publish a claim of reproducible independent evidence.

10.4 Pilot path

First show the loop on Felix and a materially different project. Only after stable authorization and failure behavior should a real organization trial be proposed, with access controlled by the organization and clear scope, baselines, security review and rollback. A possible contact or willingness to explore is not a customer, deployment, partnership or endorsement.

Measure accepted engineering outcomes, time to delivery, human interventions per task, defect escape rate, false blocks, CI-minute consumption, compute-pool usage, handoff/context cost, repair count, and time spent in qualification/landing queues. Distinguish changes in the metric instrument from changes in the product.

11. Founder attention and the control interface

The control interface should be an exception and authority console rather than a terminal manager. It should show the work that needs a decision first: hazards, blocked sessions, questions, work never reaching a PR, PRs by landing state, exhausted compute/verification pools and routing escalations. Healthy running and completed work can be collapsed. Every action should point to its owner, candidate revision, evidence, blocking reason and permitted decision.

Keep the human intervention count honest: an unrun gate and a missing test are Felix's work, not founder escalations. Permission for effects outside the reversible tree, approval of a higher-risk action, exception grants, API spending beyond policy and release authorization are genuine authority events.

11.1 Compute settings surface

Working-session model reminder (requested operating protocol). When this report is being used as the active Felix build guide, Claude should surface a one-line model recommendation in the UI before every major roadmap/section transition and before moving from architecture/research into implementation. The reminder should cite the temporary model-strength reference in Appendix D and state both the recommended model/effort and the reason.

Suggested reminder form: “MODEL REMINDER - Report prior: use Sonnet 5.5 [effort] for this bounded implementation / use Opus 5.5 [effort] for this architecture, ambiguity or authority-sensitive section. Reason: [brief reason].” This is a reminder and routing prior, not an authorization decision; the user may override it.

Setting	Policy surface
Routing policy	Adaptive sufficient-model routing | preferred provider with fallback | manual
Enabled resource pools	Per-provider toggles for subscriptions and API-backed execution
Spend controls	Never paid fallback | require approval | allow within explicit cap
Quota state	Available / constrained / exhausted / unknown, with timestamp and source
Transition evidence	Why Felix or the native agent escalated/de-escalated; what changed; what checkpoint was transferred
Trust status	Acceptance remains independent of selected model/provider; show evidence freshness and UNKNOWN obligations

The report should eventually be generated by Felix itself: an organization-status surface that distinguishes generation, qualification, landing and delivered-state throughput, plus model/compute budget, verification budget, freshness, queue age and interruption metrics. A summary limited to tokens and green checks will misdescribe the system.

12. Competitive boundary and enterprise system-of-record implications

12.1 External findings: what they establish

Finding	Source-qualified conclusion	Limit
Anthropic autonomous-agent liability	The September 29 note reports legal uncertainty around claims from autonomous agents with deep customer-system access and extended execution, including irreversible data loss or financial transactions.	Reported disclosure context; not a court ruling or independent legal opinion.
Nasdaq Calypso governed agents	Nasdaq announced an agentic environment inside Calypso with MCP integration, operational boundaries, oversight/sandboxing and access to authoritative platform records.	Primary-source product announcement; not proof of adoption volume, realized savings or willingness to pay for Felix.
OpenShell / AgentCore boundary	Existing infrastructure already covers substantial runtime policy, isolation, credentials, network mediation, policy verification, audit and generic evaluation.	Competitive scope should be rechecked before commercial absence claims.

12.2 Where Felix should and should not own the stack

Layer	Existing coverage	Felix candidate ownership
Runtime & network	OpenShell / platform security	Integrate and bind policy to task record
Identity & permission	Cloud IAM / agent policies	Task-specific authorization and delegation
Telemetry	Platform observability	Link evidence to claim and revision
Generic agent evaluation	Cloud and open evaluation stacks	Independent obligation discovery / qualification
Acceptance	Application / CI workflow	Evidence-bound human or policy decision
Cross-system state	Domain and system integrations	Authorized cross-boundary qualification
Post-delivery assurance	CI/CD and operational monitors	Selective invalidation / re-qualification

STRATEGIC BOUNDARY

The opportunity is narrower than a generic agent-control plane. Felix should own task-specific qualification, versioned obligations, independent evidence, acceptance and delivered-state invalidation. It should integrate sandboxing, identity/permissions, generic telemetry and commodity agent-evaluation substrates rather than reproduce them.

12.3 System-of-record implication

System-of-record vendors control domain semantics, permissions, authoritative data and deployment channels. Agents operating entirely inside one platform are natural incumbent features. Generic MCP connectivity, dashboards and orchestration are weak standalone differentiation. The stronger independent wedge is cross-application or cross-organization qualification: binding versioned requirements, scoped authorization and acceptance evidence across a handoff, where each party explicitly authorizes the relevant access.

12.4 Prospective industrial-test scope

The attached strategic note describes a supply-chain OS jointly developed by TSMC's own development team and a TSMC-subcontractor company, with a prospective Felix test approved in principle by the subcontractor executive contingent on internal/logistical permission and product readiness. This is not a completed pilot, TSMC endorsement, sponsorship, security/legal approval or customer status. Permission to test subcontractor-controlled work does not confer rights to joint/TSMC-controlled code, repositories, data or systems. Initial testing should use synthetic or expressly authorized scope.

13. Commercial validation experiments and decision gates

The commercial proposition remains qualified engineering throughput: higher useful autonomy while preserving inspectable evidence. Revenue, enterprise value, founder equity, dilution and personal liquidity are different quantities; the current evidence does not support numeric payoff projections.

Stage	Offer	Validation gate
1. Engineering team	Governed execution + review/evidence packet	Lower reviewer burden and fewer escaped defects
2. Enterprise workflows	Integrations + consequence-aware evidence	Repeat buyer, paid deployment, low services burden
3. Cross-system assurance	Versioned contracts across authorized boundaries	Reusable domain adapters and durable gross margin

Experiment A - Independent outcome evidence

Hypothesis: Felix's packet improves independent review over ordinary repository/CI records. Run equivalent tasks in baseline and governed workflows. Seed incomplete requirements, stale evidence and integration errors. Measure reviewer accuracy, review time, false acceptance, escaped defects and verification cost. Falsifier: no material improvement after accounting for overhead.

Experiment B - Authorized cross-organization handoff

Hypothesis: useful qualification survives a handoff without unrestricted access on either side. Use two synthetic or expressly authorized environments, with scoped requirements and evidence manifests. Measure omission detection, integrity, access-boundary violations, integration errors and review effort. Falsifier: broad access is required or ordinary handoff documents perform equally well.

Experiment C - Post-merge evidence invalidation

Hypothesis: Felix can identify when a downstream change invalidates previous acceptance. Qualify one revision, then change a dependency, integration or deployment state. Measure affected-obligation recall, false invalidation, recheck scope and cost. Falsifier: stale evidence remains accepted or trivial changes trigger disproportionate full requalification.

Experiment D - Adaptive compute economics

Hypothesis: adaptive routing can reduce total compute consumption without degrading accepted quality relative to a frontier-only baseline. Compare fixed-frontier, naive cheap-first and Felix-adaptive policies on matched tasks. Measure accepted-task success, total model usage/spend, repair count, context/handoff overhead, latency, verification cost, false escalation, false de-escalation and escaped defects. Falsifier: savings disappear after rework/verification, quality falls, or task-difficulty decisions add more overhead than they remove.

Decision summary

Decision	Disposition
Build generic sandbox or identity platform	ARCHIVE - use established enforcement substrates.
Qualification / Agent Outcome Evidence Packet	INVESTIGATE NOW - measure independent-review improvement.
Cross-organization acceptance	INVESTIGATE NOW - authorized synthetic/controlled test.
Cross-provider compute federation	BUILD NARROWLY - infrastructure behind dispatch; explicit user authorization and budgets.
Adaptive task-difficulty routing	INVESTIGATE + BENCHMARK - core Capability Plane feature, not a separate product pillar.
Deep industrial integration	WATCH - readiness, scope and permissions first.
Standalone legal-liability product	ARCHIVE FOR NOW - buyer and procurement remain unproven.

NEAR-TERM PRODUCT GATE

Do not broaden into a horizontal compliance, generic orchestration or financial-services platform. First establish a repeatable, low-friction engineering use case in which independent evidence and adaptive execution outperform ordinary agent + CI workflows on a concrete customer problem.

14. Risks, uncertainties and next measurement

Risk	Implication / test
Checker and worker circularity	Pin independent evaluator and base-ref policy; prove a candidate cannot relax its own completion criteria.
Task-difficulty misclassification	Benchmark false escalation/de-escalation; weak-model self-confidence cannot be the sole routing signal.
Cross-provider context loss	Use revision-bound checkpoints and task-scoped context; measure reconstruction cost and handoff defects.
Credential / billing boundary	Keep auth outside worker control; separate subscription and API pools; enforce explicit spend policy.
Nested compute duplication	Detect or account for host modes that internally parallelize; avoid stacking expensive orchestration layers without measured benefit.
Table parsing / malformed data	Apply schema validation and fail-closed read paths; test invalid-byte inputs and partial runs.
Local/remote revision race	Bind remote head and actual merged SHA; revalidate or refuse on concurrent movement.
CI exhaustion and candidate aging	Schedule verification capacity, cache safely and cancel superseded work.
Audit false pass	Preserve raw command exit status and pipeline semantics; measure with a stable recognition instrument.
Invisible subagent actions	Extend evidence ingestion or explicitly scope claims to parent sessions.
Machine/local environment drift	Normalize or include environmental dependencies in epochs; separate exact tree from exact runtime.
Overextension before market test	Prove one cross-plane journey; keep broad scheduler, multi-host and deployment product scope staged.

ASSESSMENT

Felix is becoming an engineering institution in architecture and operating practice, but it is not yet an independently operating one. The near-term test is not how many workers or providers it can connect. It is whether it can repeatedly deliver an authorized, independently evidenced outcome with lower human load, acceptable verification cost, and measurably better compute allocation.

Appendix A. Metrics and baseline comparison

Measure	Previous report	New frozen window	Comparison caveat
Git commits	50	32	Distinct accepted-code windows
Files changed	162	69	Diff composition changed
Additions / deletions	24,072 / 6,929	6,833 / 722	Prior handoff migration inflates churn
Test additions	6,060	3,564	Different subsystem classification
Engine additions	5,995	1,070 library/CLI	Not identical category boundaries
Raw tokens	6.975B	5.673B	Cache dominated; operating windows differ
Unique responses	30,694	25,988	Response-ID deduplicated
Tool calls	32,187	27,386	Parent + subagent
Parent sessions	190	216	Audit / scan inclusion differs
Human text turns	471	498	Excludes user-interface actions
Sessions <=1 human turn	81.6%	80.56%	Descriptive
Audit edit stops	154	88	Different exact-window and roster method
Stale + never recognized	55 / 154 (35.7%)	46 / 88 (52.3%)	Not an apples-to-apples trend
Felix stop holds	363	371	Coverage/window differs
Local suite	Median 9m56s	No completed new benchmark	Timed-out/inconclusive observations
Open PRs	Approx. 17	19 (12 drafts)	Snapshot only

New remote CI: 113 run IDs; 117 attempts; 174 job rows; 80 successful full-gate jobs (516.18 positive elapsed min), 19 failed (86.68), 1 cancelled (1.38), 17 successful reuse jobs (2.02). Positive elapsed total is 606.27 job-minutes, not an invoice. Candidate inventory: 45 worktrees, 40 distinct non-main heads, 310 unique accessible non-main commits; 173 counted since the prior baseline. These are histories and branches, not independent tasks or accepted contributions. The previous 175-commit measure used a different scope and is not directly comparable.

Adaptive execution adds no frozen-window metric here. Its evaluation begins only after the dispatch/resource instrumentation is operational and a baseline is predeclared.

Appendix B. Evidence and source register

ID	Source / purpose
B0	Felix_48h_Engineering_Systems_Report_2026-09-24_v2.pdf and DOCX: baseline measurements, methodology and earlier recommendations.
B1	Felix_Second_Engineering_Systems_Report_2026-09-27_v2.pdf: frozen-window engineering assessment merged into this edition.
B2	Felix_Strategic_Intelligence_2026-09-29.pdf: external strategic findings, competitive boundary, evidence packet and experiments.
G1	Frozen git first-parent diff and PR/worktree inspection: c6989dc5168f → 5772afac9357.
T1	Deduplicated Claude Code JSONL and parent human-text-turn scans; active telemetry to September 26 00:00 PDT.
A1	Native felix audit on temporary timestamp-filtered parent transcripts, LC_ALL=C; hold and stop classification.
C1	GitHub Actions September 24-25 gate attempts and job records; workflow source and reuse logic at frozen SHA.
S1	README.md Status and Where it is going, frozen endpoint.
S2	docs/2026-09-23-roadmap.md, Phase 0b, Capability lane, Phases 1-4.
S3	docs/2026-09-19-direction-decisions.md, especially lease and known unknowns.
S4	engine/harness/felix: gate qualification/obligations, merge boundary, route, commission, stack plan/apply.
S5	engine/harness/lib/{qualify,obligations,escape,route,ledger,adopt,commission,commission-provider,admit,stack}.sh.
S6	Frozen tracked-file and entry-point inventory for land.tsv, inventory and dispatch; absence claim limited to frozen tree.
P1	PR and CI operational snapshots: counts and timing; never interpreted as proof all tasks landed or CI minutes billed.
D1	September 28-29 design discussion: adaptive execution, native delegation first, main-agent adaptation, provider federation and task-scoped checkpoints. Proposed architecture, not frozen implementation evidence.
X6	Anthropic, Claude Sonnet 5.5, September 28, 2026 - launch benchmarks, pricing, effort behavior and positioning. https://www.anthropic.com/claude-sonnet-5-5
X7	Anthropic, Claude Opus 5.5, September 22, 2026 - flagship

ID	Source / purpose
X8	coding/agentic benchmark and pricing reference. https://www.anthropic.com/claude-opus-5-5 CodeRabbit, Claude Sonnet 5.5 for code review, September 28, 2026 - same-set hard code-review comparison against Opus 5.5. https://www.coderabbit.ai/blog/sonnet-5-5-model-review

External sources carried from the September 29 strategic note

[E1] Reuters, September 29, 2026 - Anthropic says rogue AI agents pose uncertain legal risk for the company
https://www.reuters.com/legal/litigation/anthropic-says-rogue-ai-agents-pose-uncertain-legal-risk-company-2026-09-29/

[E2] Nasdaq, September 29, 2026 - Nasdaq Calypso launches agentic capabilities across the trade lifecycle
https://www.nasdaq.com/press-release/nasdaq-calypso-launches-agentic-capabilities-scale-ai-adoption-across-trade-lifecycle

[E3] AWS Press Center, September 29, 2026 - Nasdaq Calypso launches framework to scale AI adoption
https://press.aboutamazon.com/aws/2026/9/nasdaq-calypso-launches-framework-to-scale-ai-adoption-across-the-trade-lifecycle

[E4] NVIDIA, September 28, 2026 - NVIDIA OpenShell - secure runtime for AI agents
https://www.nvidia.com/en-us/ai/openshell/

[E5] AWS, updated March 2026 - Amazon Bedrock AgentCore policy and evaluations announcement / GA updates
https://aws.amazon.com/blogs/aws/amazon-bedrock-agentcore-adds-quality-evaluations-and-policy-controls-for-deploying-trusted-ai-agents/

Source discipline. All code-based implementation claims remain limited to the frozen endpoint. External interpretive claims are hypotheses unless the source register identifies them as documented announcements/reporting. Development-plan dates are projections, not observed completion. This integrated edition does not infer new engineering progress from the September 29 strategic material.

Appendix C. Reusable report skeleton and self-generation trigger

Instruction to Felix / the report-producing model (proposed workflow, not an existing automation). Preserve the reporting method as a version-controlled repository artifact rather than relying on conversational memory. Treat the September 24 report, September 27 v2 and this September 29 integrated edition as reference examples; use the latest accepted report as the next comparison baseline.

C.1 Store the reporting contract

Create and maintain docs/reporting/engineering-systems-report-contract.md. Store the outline, typography/layout expectations, metric definitions and denominators, telemetry selectors and deduplication rules, Git and active-window conventions, source commands, comparison caveats, classification rubric, and document-generation/visual-QA procedure. Pair it with a dated measurement manifest that records each report's frozen SHA, baseline SHA, telemetry cutoff, extraction method, source revisions and evidence locations. Update these artifacts only when the method changes, recording the change and its effect on comparability.

C.2 Fixed report skeleton; variable evidence

Part	Required contents
Front matter & decision	Freeze, active interval, prior baseline, strategic-update cutoff, executive synthesis and next-cycle decision.
Measurement discipline	Sources, scope, deduplication, denominators, blind spots, changed instruments and non-comparable counts.
Engineering system	Model/tool throughput, human involvement, accepted Git/PR output, verification economics and bottlenecks.
Two-plane assessment	Capability supply, routing, adaptive execution and dispatch; Trust obligations, judge qualification, receipts and authority. Classify landed, used, partial, planned and open PR separately.
Product & lifecycle	End-to-end MVP, independent landing, post-merge/deploy evidence, observability, SLOs, migrations, rollback and pilot criteria.
Strategic / market evidence	External research, competitive boundary, commercial experiments and explicit FACT/HYPOTHESIS/UNKNOWN labeling.
Decision & audit trail	Falsifiable next tests, sequencing, resource budget, risks, metric appendix and source register.

C.3 One-command regeneration trigger

TRIGGER

“Generate the next Felix Engineering Systems Report. Read the repository reporting contract and the last accepted report. Determine the new frozen Git endpoint and a non-overlapping operational telemetry window. Collect fresh evidence through available authorized read-only tools; inspect executable code, CI and actual use, not just roadmap prose. Recalculate every metric and compare against the prior baseline with its original caveats. Re-evaluate adaptive execution using measured routing outcomes and resource pools. Re-check material external-market claims before carrying them forward. Produce the same evidence-first report in editable DOCX and PDF, render every page, and retain the measurement manifest and source register. Ask only for indispensable access, authorization or decisions; never copy old figures forward, treat unmerged work as shipped, or invent missing evidence.”

Operational limit. “Self-generate” means on request, or on a separately authorized schedule - not autonomous background execution by implication. If repository, transcript, CI or external-source access is unavailable, identify the missing sources and provide the smallest read-only collection commands or access request. Publish a qualified partial report rather than presenting absent data as zero or verified.

Appendix D. Adaptive execution design annex

D.1 Core design principles

• Two coupled routing axes: intra-host routing chooses model/effort inside the current execution environment; inter-provider routing chooses among authorized worker/provider pools and then the model/effort available there. These should converge into one allocator over execution configurations, but they may be staged separately while the product learns.

- Native delegation first: preserve the host agent's planning and subagent-selection competence; Felix adds policy, resources, evidence and authority rather than a second planner.
- Provider federation is infrastructure, not a separate product pillar. Different authorized subscriptions and API-credit pools are interchangeable resources only after capability and policy filtering.
- Difficulty recognition is the hard problem. Use execution evidence and empirical outcomes, not a worker's self-confidence alone.

- Escalation and de-escalation are both required. Hard diagnosis should not permanently pin routine implementation to frontier compute.
- Main-agent adaptation occurs at safe turn/checkpoint boundaries where supported; otherwise continue through a new worker using durable state.
- Context transfer is artifact-based: exact revision + task-scoped checkpoint + evidence, not full transcript replay or hidden-reasoning transfer.
- Trust invariance: cheaper models, subscription capacity or quota pressure never lower blocking obligations or independent evidence requirements.

D.2 Minimal adaptive flow

USER OBJECTIVE
Outcome, constraints and authority.

↓

MAIN AGENT
Native decomposition and first-pass difficulty assessment.

↓

FELIX ELIGIBILITY + RESOURCE POLICY
Allowed tools/providers, authorization, quota/spend state, task history and compute tier.

↓

WORKER EXECUTION
Native or cross-provider subagent performs bounded work in task-scoped context.

↓

TRUST PLANE
Independent checks bind evidence to revision; UNKNOWN remains blocking where required.

↓

FEEDBACK
Accept, repair, escalate, de-escalate or request authority. Outcome data updates future routing.

D.3 First benchmark matrix

Policy	Purpose	Measure
Frontier-only	Quality ceiling / cost baseline	Accepted success, latency, compute, repair, verification cost
Cheap-first	Naive cost baseline	Failure/repair amplification and total cost
Native host delegation	What Claude Code already does without Felix adaptive policy	Model mix, accepted success, intervention count
Felix adaptive	Proposed system	Same quality metrics plus routing accuracy, escalation/de-escalation precision and cross-provider overhead

The benchmark must use matched tasks and predeclared acceptance criteria. A lower token or dollar total is not a win if it increases defects, human intervention, verification load or context reconstruction enough to raise expected total cost.

D.4 Temporary Claude 5.5 model-strength reference

Research update - September 30, 2026. Sonnet 5.5 is strong enough to be the default implementation worker for well-scoped Felix work, not merely a cheap subagent. Anthropic reports Sonnet 5.5 at 70.6% on Terminal-Bench 4.0 versus 66.4% for Opus 5.5, while Opus remains ahead on FrontierCode (54.4% versus Sonnet 5.5 at 52.1% Xhigh) and CursorBench (57.8% versus 55.5%). Anthropic still describes Opus 5.5 as clearly stronger for complex, open-ended work requiring sustained judgment. Sonnet 5.5 is priced at half Opus 5.5 input/output rates and half the cache-write rate; cache-read pricing is the same. Independent CodeRabbit testing on 13 hard known-bug cases found Sonnet 5.5 caught 6/13, versus 8/13 for Opus 5.5 Standard and 10/13 for Opus 5.5 Max. Treat this as a temporary routing prior, not a permanent hard-coded policy.

Work class	Suggested Claude configuration	Felix examples / rule
Bounded routine implementation	Sonnet 5.5 Medium-High	PR cleanup, tests, docs, deterministic refactors, small bug fixes, telemetry, schema/config changes, routine provider-adapter plumbing.
Bounded multi-file implementation	Sonnet 5.5 High-Xhigh	Dispatch executor after the contract is settled, task-checkpoint serialization, provider adapters, routing telemetry, CI/evidence-cost optimizations, non-authority UI work.

Architectural or trust-boundary design	Opus 5.5 Medium-High	Dispatch contract, landing actor/candidate lifecycle, evidence graph, obligation-qualification joins, main-agent adaptation policy, cross-plane architecture.
High-ambiguity / security-critical diagnosis	Opus 5.5 Xhigh-Max	Judge independence, credential isolation, authorization boundaries, race/concurrency semantics, post-deploy invalidation, difficult root-cause work. Escalate when Sonnet evidence stalls or conflicts.

Effort is itself a routing variable. Do not reflexively run Sonnet 5.5 at Max: Anthropic reports its FrontierCode score was lower at Max than at Xhigh because extra code-review/subagent behavior sometimes caused timeouts or out-of-scope edits. For large changes with a clear architecture, a strong default pattern is Opus for architecture/acceptance review and Sonnet for implementation, with Felix Trust requirements unchanged.

Sources: Anthropic, “Claude Sonnet 5.5” (2026-09-28); Anthropic, “Claude Opus 5.5” (2026-09-22); CodeRabbit, “Claude Sonnet 5.5 for code review” (2026-09-28). External benchmarks are routing priors only; Felix should replace them with its own task -> allocation -> repair -> verification -> accepted-outcome dataset as soon as dispatch telemetry exists.

D.5 Active-build reminder protocol

Before each major build-section jump, the active Claude session should re-read the D.4 matrix and present the model reminder before proceeding. The intent is to prevent a long-lived Opus/Ultra session from becoming the default merely through inertia, while also preventing Sonnet from being used mechanically for architecture or authority-sensitive work. If a section contains mixed work, the reminder should prefer decomposition - stronger model for architecture/ambiguity, Sonnet for bounded implementation - rather than pinning the entire section to one tier.

The D.4 model guidance is explicitly temporary. Felix should replace it with its own measured routing evidence once dispatch telemetry exists. A future report should revise the reminder priors when observed task -> allocation -> repair -> verification -> accepted-outcome data contradict the external benchmark prior.

End of report.